



Burj Khalifa

An extraordinary feat of engineering, the world's tallest building and holder of several other world records



△ PIERCING THE SKIES

The Burj Khalifa dwarfs the other buildings in downtown Dubai, holding world records for the highest number of floors (163), highest occupied floor, highest observation deck, and tallest service elevator, among others.

Begun in 2004 and completed in 2009, the Burj Khalifa reaches over 2,716 ft (828 m) into the sky over Dubai. It is both the tallest building and the tallest free-standing structure in the world.

Deep foundations

The tower is built on a 12-ft- (3.7-m-) thick concrete slab supported by piles 5 ft (1.5 m) wide and 141 ft (43 m) deep. Its hexagonal central core is surrounded by a spiraling, Y-shaped plan with buttressing walls in its three arms—a design that can withstand the stresses of torsion and shearing to which the building is subjected. The building's wings decrease in size at the upper levels, a feature that breaks up the wind flow, which could cause the building to sway. The tower is clad in nearly 26,000 hand-cut reflective glass panels, designed to withstand Dubai's high temperatures, and topped by a telescopic steel spire, which was raised to its full height of more than 700 ft (213 m) by a hydraulic pump.

The Burj Khalifa is home to a hotel; restaurant; swimming pools; offices; and 900 residential apartments, including, of course, the world's highest.



△ LOOKING DOWN

The extraordinary view from the top of the tower shows the Burj Lake, the site of the 900-ft- (274-m-) long Dubai Fountain—the largest performing fountain in the world.

RECORD-BREAKING TOWER

On opening in 2010, the Burj Khalifa became the tallest building in the world, surpassing Taipei 101 in Taiwan—the previous holder of the title. It also beat the KVLV-TV mast in the US to become the tallest man-made structure in the world by some 650 ft (200 m), and beat Toronto's CN Tower (see p.53) to become the tallest free-standing structure in the world.

